

SORTING

Q7)

Question:

DS	Session	Sorting	Question Information	Level 1	Challenge 17
Question Description: Simon is studying B.Tech.-Mechanical Engineering. He's going to attend a computer science-based subject exam this semester. Due to the less preparation in the previous monthly tests, his internal mark decreased. His computer science Professor made an offer one more chance to boost up his internal marks. Professor assigns a program to Simon for the internal mark bootup. So Simon wants to solve the given task is, Given two arrays, A and B, of equal size n, the task is to find the minimum value of $A[0] * B[0] + A[1] * B[1] + \dots + A[n-1] * B[n-1]$, where shuffling of elements of arrays A and B is allowed. can you help him in solving Questions ? Constraints: $1 \leq T \leq 100$ $1 \leq N \leq 50$ $1 \leq A[i] \leq 20$					

PROGRAM:

```
1  #include <stdio.h>
2
3  int main() {
4      int T, n, i, j, temp, sum;
5      int A[50], B[50];
6
7      scanf("%d", &T);
8
9      while (T--) {
10         scanf("%d", &n);
11
12         for (i = 0; i < n; i++)
13             scanf("%d", &A[i]);
14
15         for (i = 0; i < n; i++)
16             scanf("%d", &B[i]);
17
18         for (i = 0; i < n - 1; i++) {
19             for (j = i + 1; j < n; j++) {
20                 if (A[i] > A[j]) {
21                     temp = A[i];
22                     A[i] = A[j];
23                     A[j] = temp;
24                 }
25
26                 if (B[i] < B[j]) {
27                     temp = B[i];
28                     B[i] = B[j];
29                     B[j] = temp;
30                 }
31             }
32         }
33
34         sum = 0;
35
36         for (i = 0; i < n; i++)
37             sum += A[i] * B[i];
38
39         printf("%d\n", sum);
40     }
41
42     return 0;
43 }
```

OUTPUT:

TEST CASE 1:

```
3
4
2 5 3 7
8 5 7 3
83
3
13 11 1
6 15 14
247
5
6 11 9 15 4
3 41 8 21 4
451
```

TEST CASE 2:

```
2
3
13 11 1
6 15 14 5
247
6 11 9 15 4
3 41 8 21 4
451
```